

# Output Ontology (OO)

Score: 2/5 — Significant gaps | Confidence: high

**Benchmark:** MENTALCLOUDS | **Context:** South Asia and MENA NLP/AI Practitioner Cohort — MentalCLOUDS Assessment

**Output Ontology definition:** Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

The output label space comprises (a) free-text aspect-based summaries against the three counseling components **[Q19]**, (b) the six-parameter Sekhon clinical acceptability framework — affective attitude, burden, ethicality, coherence, opportunity costs, perceived effectiveness **[Q44, Q48]**, and (c) a three-level hallucination scale **[Q46, Q53, Q54]**. For the HIGH-priority OO dimension, the label categories encode CBT-aligned Indian/US professional consensus about what counts as a clinically appropriate, coherent, or effective summary. Parameters such as 'perceived effectiveness,' 'ethicality,' and 'opportunity costs' are normative judgments that depend on the professional clinical frame applied **[Q51]**; MENA practitioners integrating Islamic psychology frameworks (TIIP, ruqyah/dhikr/salah adjuncts) **[WEB-4, WEB-5, WEB-6]** may calibrate these parameters differently. The AraHealthQA 2025 work explicitly noted that surface-level metrics 'failed to fully measure appropriateness or trustworthiness' for Arabic mental health QA **[WEB-12]**, supporting the structural concern that the OO is calibrated to a specific professional frame. The authors themselves acknowledge models score poorly on 'overall efficacy and the opportunity cost' **[Q51]** and that important negative histories (e.g., suicide risk) are not recorded **[Q64]** — a safety-critical taxonomic absence. There is no output category for religious framing, family-system salience, shame/honor-protective omissions, or fatalistic attribution.

| ID           | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| <b>Q19</b>   | "This section reports the aspect-based (psychotherapy element-based) summarization results based on the automatic evaluation scores." (p.9)                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Q44</b>   | "The evaluation framework encompasses six crucial parameters — affective attitude, burden, ethicality, coherence, opportunity costs, and perceived effectiveness." (p.12)                                                                                                                                                                                                                                                                                                                                                  |
| <b>Q46</b>   | "Additionally, we incorporate a new parameter – the extent of hallucination. It is categorical – 0 (too much hallucinated), 1 (hallucination barely observed), and 2 (no hallucination observed)." (p.12)                                                                                                                                                                                                                                                                                                                  |
| <b>Q48</b>   | "The clinical acceptability framework [39] involves six parameters – affective attitude, burden, ethicality, coherence, opportunity costs, and perceived effectiveness (c.f. Table 5 for more details)." (p.13)                                                                                                                                                                                                                                                                                                            |
| <b>Q51</b>   | "As all three models have poor scores on the more sensitive aspects i.e. the overall efficacy and the opportunity cost, this indicates that these models share the same weakness and are not suitable for clinical use as they stand now." (p.14)                                                                                                                                                                                                                                                                          |
| <b>Q53</b>   | "Additionally, the evaluation of hallucination identification is divided into three categories: no hallucination observed, hallucination barely observed, and too much hallucinated in a set of 39 conversations." (p.14)                                                                                                                                                                                                                                                                                                  |
| <b>Q54</b>   | "These categories essentially determine how well the response is consistent with the context and whether it is also incorrect, nonsensical, or contains global information beyond the scope of the conversation." (p.14)                                                                                                                                                                                                                                                                                                   |
| <b>Q64</b>   | "Important negative histories gathered during the session, such as the history of suicide risk or substance use were also not recorded, and in at least one instance, the presence of suicide risk was not identified." (p.15)                                                                                                                                                                                                                                                                                             |
| <b>WEB-4</b> | TIIP framework — Islamic psychology clinical taxonomy not represented in the annotator pool's training ( <a href="https://www.routledge.com/Applying-Islamic-Principles-to-Clinical-Mental-Health-Care-Introducing-Traditional-Islamically-Integrated-Psychotherapy/Keshavarzi-Khan-Ali-Awaad/p/book/9780367488864">https://www.routledge.com/Applying-Islamic-Principles-to-Clinical-Mental-Health-Care-Introducing-Traditional-Islamically-Integrated-Psychotherapy/Keshavarzi-Khan-Ali-Awaad/p/book/9780367488864</a> ) |
| <b>WEB-5</b> | Keshavarzi & Haque four soul aspects framework — additional MENA clinical normative reference ( <a href="https://www.tandfonline.com/doi/abs/10.1080/10508619.2012.712000">https://www.tandfonline.com/doi/abs/10.1080/10508619.2012.712000</a> )                                                                                                                                                                                                                                                                          |
| <b>WEB-6</b> | Ruqyah, dhikr, salah documented as cultural/spiritual treatment adjuncts in Muslim mental health literature ( <a href="https://pmc.ncbi.nlm.nih.gov/articles/PMC8267770/">https://pmc.ncbi.nlm.nih.gov/articles/PMC8267770/</a> )                                                                                                                                                                                                                                                                                          |

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| WEB-12 | AraHealthQA 2025 — BERTScore 'failed to fully measure appropriateness or trustworthiness' for Arabic mental health QA ( <a href="https://arxiv.org/html/2508.20047v2">https://arxiv.org/html/2508.20047v2</a> ) |
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## Strengths

- Explicit hallucination taxonomy [Q46, Q53, Q54] is largely region-neutral and transferable as a safety construct.
- Six-parameter Sekhon framework [Q44, Q48] provides a structural scaffold that, while calibrated to particular professional norms, can be partially repurposed if recalibrated by regional experts.
- Authors acknowledge clinical unsuitability of current outputs [Q51], reducing risk of overclaiming.

| ID  | Evidence                                                                                                                                                                                                                                          |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q44 | "The evaluation framework encompasses six crucial parameters — affective attitude, burden, ethicality, coherence, opportunity costs, and perceived effectiveness." (p.12)                                                                         |
| Q46 | "Additionally, we incorporate a new parameter – the extent of hallucination. It is categorical – 0 (too much hallucinated), 1 (hallucination barely observed), and 2 (no hallucination observed)." (p.12)                                         |
| Q48 | "The clinical acceptability framework [39] involves six parameters – affective attitude, burden, ethicality, coherence, opportunity costs, and perceived effectiveness (c.f. Table 5 for more details)." (p.13)                                   |
| Q51 | "As all three models have poor scores on the more sensitive aspects i.e. the overall efficacy and the opportunity cost, this indicates that these models share the same weakness and are not suitable for clinical use as they stand now." (p.14) |
| Q53 | "Additionally, the evaluation of hallucination identification is divided into three categories: no hallucination observed, hallucination barely observed, and too much hallucinated in a set of 39 conversations." (p.14)                         |
| Q54 | "These categories essentially determine how well the response is consistent with the context and whether it is also incorrect, nonsensical, or contains global information beyond the scope of the conversation." (p.14)                          |

## Checklist

### Checklist item definitions:

| ID   | Assessment Question                                          |
|------|--------------------------------------------------------------|
| OO-1 | Review output label categories for regional relevance        |
| OO-2 | Identify missing regionally specific categories              |
| OO-3 | Flag categories encoding non-regional values                 |
| OO-4 | Consider stakeholder-driven taxonomy redesign                |
| OO-5 | Document taxonomy issues harming structural/content validity |

### Checklist responses:

**OO-1:** The six clinical-acceptability parameters [Q44] are partially region-relevant but encode CBT-aligned professional norms; their calibration in MENA Islamic-integrated practice may differ [WEB-4, WEB-5, WEB-6]. The free-text summary output [Q19] is region-agnostic at the form level but scored against region-specific reference summaries.

**OO-2:** Missing categories include: religious-framing salience, shame/honor-protective omission detection, family-system content, fatalistic attribution flags, psychotherapy-type identification [Q63], and reliable suicide-risk and negative-history capture [Q64]. All are clinically salient for the SA-MENA target context.

**OO-3:** 'Perceived effectiveness' and 'opportunity costs' [Q44, Q51] encode professional-frame-dependent value judgments that may not align with MENA clinical norms integrating Islamic psychology [WEB-4, WEB-5]. 'Ethicality' likewise depends on the underlying clinical-ethical framework (e.g., Qatar's CILE Islamic bioethics integration [WEB-20]) which differs from the unstated frame in the benchmark.

**OO-4:** Stakeholder-driven taxonomy redesign is warranted given HIGH OO priority: MENA clinical psychologists and Islamic psychology researchers should be involved in extending the parameter set and recalibrating the existing six [WEB-4, WEB-16].

**OO-5:** Documented taxonomy issues harming structural and content validity: (a) missing region-relevant categories (religion, family, shame, fatalism), (b) value-laden parameters calibrated to undocumented professional norms, (c) safety-critical gap on

suicide-risk capture [Q64] compounded by regional stigma [WEB-1, WEB-2, WEB-3], (d) absence of NLP-operationalized Islamic psychology categories [WEB-4].

| ID     | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| Q19    | "This section reports the aspect-based (psychotherapy element-based) summarization results based on the automatic evaluation scores." (p.9)                                                                                                                                                                                                                                                                                                                                                                                |
| Q44    | "The evaluation framework encompasses six crucial parameters — affective attitude, burden, ethicality, coherence, opportunity costs, and perceived effectiveness." (p.12)                                                                                                                                                                                                                                                                                                                                                  |
| Q51    | "As all three models have poor scores on the more sensitive aspects i.e. the overall efficacy and the opportunity cost, this indicates that these models share the same weakness and are not suitable for clinical use as they stand now." (p.14)                                                                                                                                                                                                                                                                          |
| Q63    | "The models are also unable to identify psychotherapy types (e.g., cognitive behavior therapy) and therapy techniques, which form an integral part of counseling notes." (p.15)                                                                                                                                                                                                                                                                                                                                            |
| Q64    | "Important negative histories gathered during the session, such as the history of suicide risk or substance use were also not recorded, and in at least one instance, the presence of suicide risk was not identified." (p.15)                                                                                                                                                                                                                                                                                             |
| WEB-1  | BMC Psychiatry 2020 systematic review (n=6,767) — stigma toward mental illness in India higher than Western countries ( <a href="https://link.springer.com/article/10.1186/s12888-020-02937-x">https://link.springer.com/article/10.1186/s12888-020-02937-x</a> )                                                                                                                                                                                                                                                          |
| WEB-2  | Pakistan NPMS 2022 — ~1 in 3 adults meet psychiatric criteria; help-seeking low partly due to stigma ( <a href="https://www.medrxiv.org/content/10.1101/2024.07.08.24310056v1.full.pdf">https://www.medrxiv.org/content/10.1101/2024.07.08.24310056v1.full.pdf</a> )                                                                                                                                                                                                                                                       |
| WEB-3  | Bangladesh 2023 — mental health treatment gap ~94%; stigma a primary structural barrier ( <a href="https://pmc.ncbi.nlm.nih.gov/articles/PMC10579681/">https://pmc.ncbi.nlm.nih.gov/articles/PMC10579681/</a> )                                                                                                                                                                                                                                                                                                            |
| WEB-4  | TIIP framework — Islamic psychology clinical taxonomy not represented in the annotator pool's training ( <a href="https://www.routledge.com/Applying-Islamic-Principles-to-Clinical-Mental-Health-Care-Introducing-Traditional-Islamically-Integrated-Psychotherapy/Keshavarzi-Khan-Ali-Awaad/p/book/9780367488864">https://www.routledge.com/Applying-Islamic-Principles-to-Clinical-Mental-Health-Care-Introducing-Traditional-Islamically-Integrated-Psychotherapy/Keshavarzi-Khan-Ali-Awaad/p/book/9780367488864</a> ) |
| WEB-5  | Keshavarzi & Haque four soul aspects framework — additional MENA clinical normative reference ( <a href="https://www.tandfonline.com/doi/abs/10.1080/10508619.2012.712000">https://www.tandfonline.com/doi/abs/10.1080/10508619.2012.712000</a> )                                                                                                                                                                                                                                                                          |
| WEB-6  | Ruqyah, dhikr, salah documented as cultural/spiritual treatment adjuncts in Muslim mental health literature ( <a href="https://pmc.ncbi.nlm.nih.gov/articles/PMC8267770/">https://pmc.ncbi.nlm.nih.gov/articles/PMC8267770/</a> )                                                                                                                                                                                                                                                                                          |
| WEB-16 | Stanford Muslim Mental Health & Islamic Psychology Lab — institutional locus for Islamic psychology NLP ( <a href="https://med.stanford.edu/mmhip.html">https://med.stanford.edu/mmhip.html</a> )                                                                                                                                                                                                                                                                                                                          |
| WEB-20 | Qatar Ministry of Public Health collaborates with CILE on Islamic bioethics frameworks for healthcare AI ( <a href="https://pmc.ncbi.nlm.nih.gov/articles/PMC12141507/">https://pmc.ncbi.nlm.nih.gov/articles/PMC12141507/</a> )                                                                                                                                                                                                                                                                                           |

## Information Gaps

- The exact operationalization of each Sekhon parameter for the annotation task (rubric examples for 0, 1, 2) is not surfaced in the available quotes; this affects the precision of cross-cultural calibration assessment.

## Requires Expert Verification

- MENA clinical psychologists with Islamic psychology training should review the six Sekhon parameters and propose recalibrations or additions for regional applicability.

## Remediation

**Gap 1: Six-parameter Sekhon framework [Q44, Q48] encodes professional-norm-dependent value judgments (perceived effectiveness, ethicality, opportunity costs) calibrated to undocumented Indian/US frames; no categories for religious-framing salience, shame-protective omission, or family-system content; safety-critical suicide-risk capture is absent [Q64].**

**Recommendation:** Convene a MENA + South Asian expert panel including TIIP-trained clinicians [WEB-4, WEB-5] and Qatar-CILE-style Islamic bioethics specialists [WEB-20] to recalibrate the six parameters, add region-specific categories, and define a safety-critical suicide-risk-capture sub-rubric.

| ID | Evidence |
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| <b>Q44</b>    | "The evaluation framework encompasses six crucial parameters — affective attitude, burden, ethicality, coherence, opportunity costs, and perceived effectiveness." (p.12)                                                                                                                                                                                                                                                                                                                                                  |
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| <b>WEB-20</b> | Qatar Ministry of Public Health collaborates with CILE on Islamic bioethics frameworks for healthcare AI ( <a href="https://pmc.ncbi.nlm.nih.gov/articles/PMC12141507/">https://pmc.ncbi.nlm.nih.gov/articles/PMC12141507/</a> )                                                                                                                                                                                                                                                                                           |